

Javascript

Theory

1. Session Storage

- **Definition:** Stores data for a single browser session. Cleared when the tab/window is closed.
- **Methods:**
 - `setItem(key, value)` → store data.
 - `getItem(key)` → retrieve data.
 - `removeItem(key)` → delete one item.
 - `clear()` → delete all items.
- **Difference from LocalStorage:** LocalStorage persists even after closing the browser, while SessionStorage does not.

2. Cookies

- **Definition:** Small pieces of data stored in the browser, often used for authentication and tracking.
- **Syntax:**

```
document.cookie = "username=Raj";
```

- **Limitations:** Size limit (~4KB), sent with every HTTP request.

3. Math Object

- Provides mathematical operations.
- Common methods:
 - `Math.ceil(x)` → round up.
 - `Math.floor(x)` → round down.
 - `Math.max(a, b, c)` → largest value.
 - `Math.min(a, b, c)` → smallest value.
 - `Math.abs(x)` → absolute value.
 - `Math.pow(a, b)` → exponentiation.
 - `Math.random()` → random number between 0 and 1.

4. Async Fetch API

- **Definition:** Used to make HTTP requests asynchronously.
- **Syntax:**

```
async function fetchData() {  
  
  const res = await fetch("https://api.example.com");  
  
  const data = await res.json();  
  
  console.log(data);  
  
}
```

- **Error Handling:** Use try...catch to handle failures.
- **Event Listeners:** addEventListener("click", callback) can trigger fetch calls dynamically.

Coding Round Questions

- Session storage → Explain SessionStorage with methods and differences from LocalStorage.
- Cookies → How do cookies differ from SessionStorage?
- Math object → Show usage of ceil, floor, max, min, abs, pow, and random.
- Fetch API → Write a program to fetch data from an API using async/await.
- Form validation → Write a function to validate form input before submission.
- Event listeners → Show how to attach a click event to a button.

Machine Coding Round Challenge

Problem: Joke Generator with Session Tracking

Requirements:

- Create a form to input user name.
- Store the name in **SessionStorage**.
- Display jokes fetched from <https://icanhazdadjoke.com/slack>.
- Add a **Next Joke** button to fetch new jokes.
- Use **Math.random()** to simulate random joke selection.
- Store last fetched joke in **cookies** for persistence.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Session</title>
  </head>
  <body>
    <h2>Session storage</h2>

    <form onsubmit="return validateForm();">
      <input type="text" id="name" placeholder="Enter your name" />

      <button type="submit">Submit</button>
    </form>

    <p id="content"></p>

    <button id="joke">Next joke</button>

    <script>
      const content = document.getElementById("content");
      const button = document.getElementById("joke");
      function validateForm() {
        const nameField = document.getElementById("name").value;

        if (nameField === "") {
          alert("Name field is required");
          return false;
        }

        alert("Form submitted");
        return true;
      }

      sessionStorage.setItem("name", "Fiit");

      console.log(sessionStorage.getItem("name"));

      sessionStorage.removeItem("name");

      sessionStorage.clear();

      document.cookie = "username = Raj";

      console.log(document.cookie);

      console.log(Math.ceil(5.3));
```

```
console.log(Math.floor(5.6));

console.log(Math.max(10, 20));

console.log(Math.min(20, 40, 10));

console.log(Math.abs(-20));

console.log(Math.pow(3, 3));

console.log(Math.random(10, 20, 40));

async function fetchData() {
  try {
    const result = await fetch("https://icanhazdadjoke.com/slack");

    console.log(result);

    const data = await result.json();
    console.log(data);

    console.log(data.attachments[0].text);
    content.textContent = data.attachments[0].text;
  } catch (error) {
    console.log(error.message);
  }
}

button.addEventListener("click", () => {
  fetchData();
});
</script>
</body>
</html>
```